

SUPERALLOYS

A Technical Guide

Second Edition

Matthew J. Donachie
Stephen J. Donachie



**The Materials
Information Society**

Materials Park, OH 44073-0002
www.asminternational.org

Copyright © 2002
by
ASM International®
All rights reserved

No part of this book may be reproduced, stored in a retrieval system, or transmitted, in any form or by any means, electronic, mechanical, photocopying, recording, or otherwise, without the written permission of the copyright owner.

First printing, March 2002

Great care is taken in the compilation and production of this book, but it should be made clear that NO WARRANTIES, EXPRESS OR IMPLIED, INCLUDING, WITHOUT LIMITATION, WARRANTIES OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE, ARE GIVEN IN CONNECTION WITH THIS PUBLICATION. Although this information is believed to be accurate by ASM, ASM cannot guarantee that favorable results will be obtained from the use of this publication alone. This publication is intended for use by persons having technical skill, at their sole discretion and risk. Since the conditions of product or material use are outside of ASM's control, ASM assumes no liability or obligation in connection with any use of this information. No claim of any kind, whether as to products or information in this publication, and whether or not based on negligence, shall be greater in amount than the purchase price of this product or publication in respect of which damages are claimed. THE REMEDY HEREBY PROVIDED SHALL BE THE EXCLUSIVE AND SOLE REMEDY OF BUYER, AND IN NO EVENT SHALL EITHER PARTY BE LIABLE FOR SPECIAL, INDIRECT OR CONSEQUENTIAL DAMAGES WHETHER OR NOT CAUSED BY OR RESULTING FROM THE NEGLIGENCE OF SUCH PARTY. As with any material, evaluation of the material under end-use conditions prior to specification is essential. Therefore, specific testing under actual conditions is recommended.

Nothing contained in this book shall be construed as a grant of any right of manufacture, sale, use, or reproduction, in connection with any method, process, apparatus, product, composition, or system, whether or not covered by letters patent, copyright, or trademark, and nothing contained in this book shall be construed as a defense against any alleged infringement of letters patent, copyright, or trademark, or as a defense against liability for such infringement.

Comments, criticisms, and suggestions are invited, and should be forwarded to ASM International.

Prepared under the direction of the ASM International Technical Books Committee (2001–2002), Charles A. Parker, Chair

ASM International staff who worked on this project included Veronica Flint, Manager of Book Acquisitions; Bonnie Sanders, Manager of Production; Jill Kinson, Production Project Manager; and Scott Henry, Assistant Director of Reference Publications.

Library of Congress Cataloging-in-Publication Data

Donachie, Matthew J.

Superalloys : a technical guide / M. Donachie, Jr., S. Donachie.—2nd ed.
p. cm.

Includes bibliographical references and index.

ISBN 0-87170-749-7

1. Heat resistant alloys. I. Donachie, S. (Steve) II. Title.

TN700 .D66 2002

620.1'617—dc21

2001055227

ISBN: 0-87170-749-7

SAN: 204-7586

ASM International®

Materials Park, OH 44073-0002

www.asminternational.org

Printed in the United States of America

Contents

Dedication.....	vii
Preface.....	ix
Chapter 1: Superalloys for High Temperatures—a Primer.....	1
How and When to Use This Chapter.....	1
Some History.....	1
What Are Superalloys and What Can You Do to Them?.....	2
A Short Review of the High-Temperature Strength of Metals.....	2
Basic Metallurgy of Superalloys	2
Some Superalloy Characteristics and Facts.....	8
Applications.....	8
What to Look for in This Book	9
Chapter 2: Selection of Superalloys	11
Overview.....	11
Wrought versus Cast Superalloys	15
The Properties of Superalloys	18
Selecting Superalloys	22
Chapter 3: Understanding Superalloy Metallurgy	25
Groups, Crystal Structures, and Phases	25
Introduction to the Alloy Groups.....	26
Alloy Elements and Microstructural Effects in Superalloys	29
Microstructure.....	30
Superalloy Strengthening.....	32
Function of Processing in Microstructure Development	38
Chapter 4: Melting and Conversion.....	42
Solidification of Superalloys	42
Electric Arc Furnace (EAF)/Argon Oxygen Decarburization (AOD) Overview	44
Electric Arc Furnace/Argon Oxygen Decarburization Operation.....	46
Vacuum Induction Melting (VIM) Overview	50
Vacuum Induction Melting Operation	51
Consumable Remelt Overview.....	56
Electrode Quality	58
Vacuum Arc Remelting Operation	58
Melt-Related Defects in VAR	64
Electroslag Remelting Operation	66
Melt-Related Defects in ESR.....	71
Triple-Melted Products.....	71
Ingot Conversion and Mill Products	72

Chapter 5: Investment Casting	79
Introduction	79
Investment Casting Practice	80
Investment-Cast Components	84
Investment Casting Problems	85
Superalloy Castings	89
Chapter 6: Forging and Forming.....	91
Forging and Related Processes.....	91
Forging Basics	92
Forging Considerations	93
The Forging Process	94
Practical Forging Considerations	101
Forming of Superalloys.....	106
Practical Forming of Superalloys.....	108
Formability Processes	110
Some Additional Aspects of Forming: Iron-Nickel and Nickel-Base Superalloys	111
Some Additional Aspects of Forming: Carbide-Hardened Cobalt-Base Superalloys	113
Superplastic Forming/Forging	113
Chapter 7: Powder Metallurgy Processing	117
Powder Superalloys Overview	117
Powder Metallurgy Powder Production Techniques	120
Powder Metallurgy Powder Consolidation Techniques	124
Powder-Based Disk Components.....	125
Other Powder-Based Superalloy Components	129
Chapter 8: Heat Treating	135
Introduction	135
Heat Treatment Types	137
Heat Treatment Procedures	139
Surface Attack and Contamination.....	142
Protective Atmospheres	143
Furnace Equipment.....	144
Practical Heat Treatment of Superalloys.....	145
Chapter 9: Joining Technology and Practice	149
Introduction	149
Joining the Alloy Classes	150
Joint Integrity and Design	151
Cracking and Soundness of Fusion-Welded Superalloys.....	152
Preweld and Postweld Heat Treatments for Fusion Welding	160
Welding Specifications	161
Fusion Welding Practice for Superalloys	161
Practical Aspects of Superalloy Fusion Welding.....	163
Superalloy Fusion Welding Details.....	165
Superalloy Solid-State Joining	173
Superplastic Forming/Bonding of Components	175
Brazing	175
Brazing Processes.....	178
Brazing Superalloys.....	181
Transient Liquid Phase (TLP, Pratt & Whitney) Bonding.....	183
Some Superalloy Joining Illustrations	186

Chapter 10: Machining.....	189
Introduction	189
Overview of Superalloy Machining	189
Specific Machining Operations.....	194
Chapter 11: Cleaning and Finishing.....	203
Introduction	203
Metallic Contamination Removal	204
Tarnish Removal	205
Oxide and Scale Removal	205
Finishing Processes.....	209
Cleaning and Finishing Problems and Solutions.....	210
Chapter 12: Structure/Property Relationships	211
Introduction	211
General Aspects of Precipitation Hardening in Superalloys	213
Grain-Boundary Carbides in Nickel-Base Superalloys	218
Grain-Boundary Carbides in Other Superalloys.....	222
Carbide Precipitation—General Hardening.....	222
IN-718 and the Role of δ Phase in Strengthening	225
Cast and Wrought Superalloy Commentary	226
Wrought Superalloys—Physical, Tensile, and Creep-Rupture Properties.....	241
Wrought Superalloys—Fatigue and Fracture Properties	250
Cast Superalloys—Physical, Tensile, and Creep-Rupture Properties.....	258
Chapter 13: Corrosion and Protection of Superalloys.....	287
Overview.....	287
Oxidation/Corrosion Testing of Superalloys and Their Coatings	289
Degradation by Gaseous Oxidation or Mixed Gases.....	294
Hot Corrosion	298
Coatings for Superalloy Protection	309
Diffused Aluminide Coatings	311
Overlay Coatings.....	316
Thermal Barrier Coatings	319
Coating Comparisons.....	321
Chapter 14: Failure and Refurbishment.....	323
Overview.....	323
Overheating and Microstructural Stability	324
Microstructural Degradation.....	327
Failures of Superalloy Components	330
Damage Recovery, Refurbishment, and Repair	334
Chapter 15: Superalloys—Retrospect and Future Prospects.....	339
The 20th Century	339
The 21st Century.....	345
Appendix A: Source Information	353
Some Superalloy Information/Product Sources.....	353
Sources for Collected Property Data on Superalloys.....	354
Appendix B: Some Additional Microstructural Information	357
Introduction	357
Topological Close-Packed Phase Formation.....	357
Appendix C: Other Sources.....	365
Subject Index.....	371
Alloy Index	409

Dedication

We wish to dedicate this book to our parents, Viola and Matthew Donachie, and our wives, Cynthia and Martha.

Our father was an outstanding self-taught metallurgist. He worked as an electrician in the steel mills in Motherwell, Scotland, for a short time before emigrating to the United States. “Scottie” to his friends, “Steve” to his family, and “Matthew” to later life acquaintances only, was a long time ASMer and introduced us to the art and science of the metallurgy field. Many an hour was spent by us at Steve’s labs in Holyoke, MA as we grew through elementary, junior high, and high school. Metallography was an art to be learned at the master’s knee. Photography was a passion. All sorts of improvisations were made in the lab to produce the most wondrous metallurgical results. A patient and thorough man, Steve was most responsible for each of us in turn to choose to become a metallurgist and, eventually, to go on to receive our doctorates in the field.

Vi was responsible for our education for a many a year because Steve was away for weeks at a time during the war years of the 1940s. She made certain that we accomplished our studies and encouraged us at all times. Little did we realize the depths of her own talent until we finally were off to college and discovered that Vi had become a painter on canvas, an architect of elegant enamelware, a ceramist, a weaver of some note, and an occasional judge at competitions. She presided over the Holyoke Woman’s Club and, at another time, over the Home Information Center. Yes, we remembered that she crocheted and occasionally knitted when we were younger, and her avid reading and breadth of knowledge of current affairs were remarkable. Still, Vi submerged most of her talents until later years to bring all her children to college and beyond. She was quite a remarkable woman!

The patience and character of many women are legendary, and we would like to hold out the examples of our wives, Cynthia (Steve’s wife) and Martha (Matt’s wife), who have put up with the workaholic nature of our lives for decades. As we have worked at various tasks over the years, they understood and helped to make it easier for us to complete those tasks. Many an evening was spent without our presence, yet they both have encouraged us. It is to them that we also dedicate this book.

With the help and encouragement of our parents and wives, we learned and, hopefully, practiced intellectual investigation and ethical exploration and exposition of knowledge in our chosen areas of metallurgy. We are truly indebted to all for the ways they have influenced our lives.

“How happy is he born and taught
That serveth not another’s will
Whose armour is his honest thought
And simple truth his utmost skill!”

Sir H. Wotton in *The Golden Treasury*, F. T. Palgrave, 1861

Steve
Matt

Preface

The superalloys are, indeed, super. For over 6 decades now, they have provided the most reliable and cost effective means of achieving high operating temperature and stress conditions in aircraft and, now, industrial gas turbines. They have resisted all efforts to reduce their importance and decrease the volume of use. Instead, superalloys continue in wide use in the gas turbine field and may well begin to see even more volume of use in other fields. Superalloys now find application in such diverse fields as oil equipment and biomedical implants. As we move through the first decade of the twenty-first century, superalloys seem secure. To be sure, advances in alloy chemistry are not so easy to achieve any more, but it is being done. Surface modification, partly through the application of coating technology, has extended the useful temperature range of alloys concurrent with the introduction of directional structures and then single crystals of superalloys. Melting technology is “head and shoulders” above that of just 15 years ago!

In the late 40s and 50s, there were some conferences and a few published books catering to the developing field of superalloys. At Special Metals, a new generation of processing was dawning as vacuum melting of commercial alloys became a reality. By the mid 60s, the majority of the alloys in use today, except for the directionally solidified ones, existed. The 60s saw the zenith of superalloy development as columnar grain alloys and single crystals were made feasible, and many polycrystalline alloys were brought to commercial reality. Papers on superalloys at the ASM and AIME meetings became fairly routine. At the end of the decade, an important conference was set into being by a dedicated group of metallurgists representing ASM, AIME, and ASME. The first International Conference on Superalloys was not originally intended to be the nucleus of a long running forum, but it did indeed become that. The conference, known as the Seven Springs Conference after the original and only conference location, has continued from 1968 into the twenty-first century. Some other conferences have been initiated and prospered as well. Some conferences cover only specific alloys; e.g. Inconel 718 and related alloys are the subjects of a continuing series of conference.

ASM was an early leader in the presentation of books on high-temperature behavior of metals. In 1979, ASM published *Source Book on Materials at Elevated Temperatures*, in 1984, the *Superalloys Source Book*, and in 1988, the first edition of *Superalloys: A Technical Guide*. Other books on high temperature behavior/properties have been published as well by ASM. The continued success of superalloy technology has encouraged us to undertake a total revision of *Superalloys: A Technical Guide*. The new Second Edition contains much more information than the previous edition and has been modified in layout to better accommodate the technical information provided. The text has been completely revised and expanded from that of the previous edition with many additional figures and new and revised tables.

Virtually all technical aspects of superalloys are covered in this edition. The book is not intended to be exhaustive in every respect, but we believe that the

reader will find it to be most comprehensive. Chapter 4 in particular is probably the most complete and up-to-date presentation on alloy melting available. Selection of alloys is covered with many suggestions to lead the reader to ask appropriate questions either of her/himself or others in the application or development of superalloys. Furthermore, the relation of properties and microstructure is covered in more detail than in previous books. The Guide has been reviewed for accuracy, but it is possible that errors will have occurred. The writers would appreciate receiving either corrections or suggestions (or both) from readers.

If you are new to the use of superalloys, we would strongly suggest starting with Chapter 1. “Superalloys for High Temperatures—A Primer” will suit the needs of readers who want just a brief introduction to superalloys and cannot spend more time on the subject. If you are knowledgeable in metallurgy but have limited knowledge of superalloys, you might wish to start with Chapter 3, “Understanding Superalloy Metallurgy,” before proceeding to one of the specialized chapters for more in-depth information. It is most likely that your immediate needs can be satisfied by perusing this book. However, on completing appropriate chapters, you may wish to pursue reading from one of the references listed in Appendix B.

The writers wish to thank all those who contributed to this book, including the many contributors to other ASM books and the *ASM Handbook* series. We extend our special thanks to John Marcin and Joe Goebel who extensively reviewed Chapters 5 and 13 respectively. This book is the product of the authors’ experience in superalloys, totaling close to 60 years between them, the authors’ personal biases, their technical files, and the extensive resources of ASM International. We particularly would like to thank Veronica Flint, retired from ASM International, for her encouragement to pursue this work and for her perseverance over the several years as the material made its way into electronic and now hard copy form. Veronica Flint and Matt have worked on several past ASM books. It was always been a pleasure to work with Veronica and was especially so on this significant update of an important technical field. The successful publication of this Second Edition is a tribute once more to the dedication of ASM International to providing the greatest access to materials information for the widest possible audience.

MJD

mattd@monad.net

SJD

sdonachie@specialmetals.com

October 2001



ASM International is the society for materials engineers and scientists, a worldwide network dedicated to advancing industry, technology, and applications of metals and materials.

ASM International, Materials Park, Ohio, USA
www.asminternational.org

This publication is copyright © ASM International®. All rights reserved.

Publication title	Product code
Superalloys: A Technical Guide	06128G

To order products from ASM International:

Online Visit www.asminternational.org/bookstore

Telephone 1-800-336-5152 (US) or 1-440-338-5151 (Outside US)

Fax 1-440-338-4634

Mail Customer Service, ASM International
9639 Kinsman Rd, Materials Park, Ohio 44073-0002, USA

Email CustomerService@asminternational.org

In Europe American Technical Publishers Ltd.
27-29 Knowl Piece, Wilbury Way, Hitchin Hertfordshire SG4 0SX,
United Kingdom
Telephone: 01462 437933 (account holders), 01462 431525 (credit card)
www.ameritech.co.uk

In Japan Neutrino Inc.
Takahashi Bldg., 44-3 Fuda 1-chome, Chofu-Shi, Tokyo 182 Japan
Telephone: 81 (0) 424 84 5550

Terms of Use. This publication is being made available in PDF format as a benefit to members and customers of ASM International. You may download and print a copy of this publication for your personal use only. Other use and distribution is prohibited without the express written permission of ASM International.

No warranties, express or implied, including, without limitation, warranties of merchantability or fitness for a particular purpose, are given in connection with this publication. Although this information is believed to be accurate by ASM, ASM cannot guarantee that favorable results will be obtained from the use of this publication alone. This publication is intended for use by persons having technical skill, at their sole discretion and risk. Since the conditions of product or material use are outside of ASM's control, ASM assumes no liability or obligation in connection with any use of this information. As with any material, evaluation of the material under end-use conditions prior to specification is essential. Therefore, specific testing under actual conditions is recommended.

Nothing contained in this publication shall be construed as a grant of any right of manufacture, sale, use, or reproduction, in connection with any method, process, apparatus, product, composition, or system, whether or not covered by letters patent, copyright, or trademark, and nothing contained in this publication shall be construed as a defense against any alleged infringement of letters patent, copyright, or trademark, or as a defense against liability for such infringement.